ITE 3962 – Final Year Project

Project Proposal

Garbage Collection Management System

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Introduction

Waste generation has been significantly increasing throughout the past decade, and is expected to increase even more rapidly in the future due to a lack of modern infrastructure, rapid urbanization, and population growth. These will lead to serious problems such as health problems, unsanitary living conditions, economic losses, and more. Statistics show that annual waste generation is expected to increase from 70% by 2050, which is 3.4 billion tons, and average solid waste generation per day is 7000 MT, and the Colombo municipal council accounts for 0.8kg of garbage per head per day.[1]

The traditional method of garbage collection is used in many places in Sri Lanka, garbage is collected without a systemic process often lacks transparency, a lack of coordination between the municipal council and residents, and late or missed pickups (Irregular waste collections), unmanaged wastes (overflowing garbage cans), and people's improper knowledge regarding garbage management & garbage type. These issues are serious for creating a peaceful and healthy living environment.

To handle these issues, this project proposes a technology-driven, web-based garbage collection management system, grounded in research. Although some mobile apps act as garbage management systems for Sri Lanka, they fall short in addressing garbage collectors and residents. The proposed platform will enable real-time tracking of garbage trucks, efficient management of drivers and collectors, and a proper system for handling complaints from residents, and enable residents to detect the garbage type. It will aim to promote a cleaner, more organized, and garbage-free Sri Lanka.

In addition, the system will include features such as resident registration, daily garbage collection scheduling, and event planning for awareness programs. Also, it is designed to provide day-to-day operations and public garbage management. Manual processes often result in human errors, data loss, and miscommunication, which can impact overall efficiency. Also, planning to add a quick response code on bins to identify the type of garbage. This system will help to reduce residents' complaints and improve public satisfaction. Overall, the Garbage Management System will reduce public complaints, improve satisfaction, and support sustainable waste management practices across communities.

Background & Motivation

Garbage collection has become one of the most serious problems in Sri Lanka. There are many health problems, such as cholera, typhoid, malaria, and dengue fever, because of improper garbage disposal. These health problems can be fatal and are often the result of uncleaned conditions caused by uncollected or improperly dumped garbage. Unclean garbage places attract flies, mosquitoes, and rats, posing a direct threat to residents' health.

One of the main contributors to this issue is the unstructured and inconsistent waste collection process carried out by municipal councils. There is no fixed schedule for garbage trucks to collect waste in many areas (almost all areas). Residents often place garbage too early, leading to overflowing bins before collection, or too late, causing waste to remain unattended for days. These inefficiencies result in garbage piling up in public spaces, spreading foul odors, and attracting disease-carrying animals.

And also, the current garbage collection system lacks coordination, transparency, and community involvement. There is no way for residents to track the garbage collection lorry or receive notification. And residents are not aware of garbage separation, they put all types of garbage together in one bag and dispose in one bin, no way for residents to file complaints or contact officers regarding garbage collection problems. This disconnection leads to poor garbage disposal in areas.

Motivated by these problems, backgrounds, and impacts on residents' health and the environment, this project is introducing a technology-driven solution. As an IT student, I am creating a platform that can play the role of a connector between residents and the municipal council, automate the key tasks, and reduce human errors. The proposed system will contribute significantly toward cleaner surroundings and improved quality of life

Problem in Brief

The current garbage collection management system is poorly coordinated, with an irregular collection schedule and a lack of transparency. Residents are often unaware of the garbage pickup schedule, while the municipal council lacks real-time information about the status of garbage bins. Currently, there is no proper tracking system in place, so neither residents nor the municipal council can know when the garbage truck starts& its route, where it is currently located, or whether the bins have been emptied. Also, residents are unaware of garbage type so when dumping it creates huge harm to the environment.

There is no efficient way to receive notifications or file complaints about drivers or bins. Residents have to wait for the truck without the knowledge of its arrival, and Residents need to visit the municipal council office and have to file a complaint manually through letters. It can create human errors and minimise efficiency.

These gaps in communication and system design are resulting in increased health risks, environmental harm, and public frustration—highlighting the urgent need for a technology-driven, user-friendly solution.

Aim and Objective

Aim

The aim of this project is to develop a system for addressing the issue of garbage collection. Planning to create a system that enhances communication between residents and the municipal council, enables real-time tracking of the status of garbage trucks for the residents and municipal council, and allows residents to get knowledge of the garbage type by uploading a picture to the website.

Objectives

- Initiate an efficient garbage collection system by evading infirmities in the traditional system.
- Increase the contribution and involvement of the public in the project.
- Research and evaluate requirements related to key aspects of garbage management, such as bin placement, transport logistics, and garbage separation.
- Create a web application tailored to residents' needs and conduct a formal evaluation to ensure its effectiveness and usability.

Proposed Solution

The proposed solution is a web-based garbage collection and management platform designed to simplify and enhance the user experience for residents looking for proper garbage disposal. This platform will allow residents to register for getting service, and drivers and officers to register for providing service. Through the website, residents can upload images of their garbage and gain knowledge about their garbage type, and then they can dispose of it in the proper bin. Residents will be able to track the garbage truck and receive real-time updates, view garbage truck collection schedules, report issues like missed pickups or overflowing bins, get information, and update about garbage collection. Also, this platform will support municipal council staff (admin) to manage daily operations such as trekking collection reports, assigning routes, assigning drivers, receiving complaints, taking actions on complaints, creating/ updating new bins, assigning drivers & trucks. Also, drivers can upgrade their work/bin status. Residents, drivers, and admins can also create profiles and update them. The system will also promote public awareness by sharing recycling tips, event updates, and health guidelines, thereby encouraging responsible waste disposal and environmental sustainability.

By offering a comprehensive and user-friendly interface, the website will cater to residents and municipal staff seeking convenience, efficiency, and a streamlined waste management experience.

Reference

[1] Dasuni Nawinna and Anuradha Karunasena, "A Smart System for Solid Waste Management in Sri Lanka," *Conf. 9th Int. Conf. Sustain. Built Environ. 2018At Kandy, Sri Lanka*, no. 2018.

Appendix A: Plan of action

Task		Time Period									
		01	02	03	04	05	06	07	08	09	10
1	**Requirements Analysis and Definition**										
1.1	Define Requirements										
1.2	Define scope and risk management										
1.3	Develop the project proposal										
2	**System and Software Design**										
2.1	Identify the best solution										
2.2	Implement it										
3	**Implementation and Unit Testing**										
3.1	User layout design										
3.2	Database design										
3.3	Other requirements										
4	**Integration and Unit Testing**										
4.1	Develop programming										
4.2	Develop the functionality										
4.3	Content integration										
5	**Operation and Maintenance**										

5.1	Testing with different devices										
5.2	Test and debug										

Appendix 1: Plan of action